

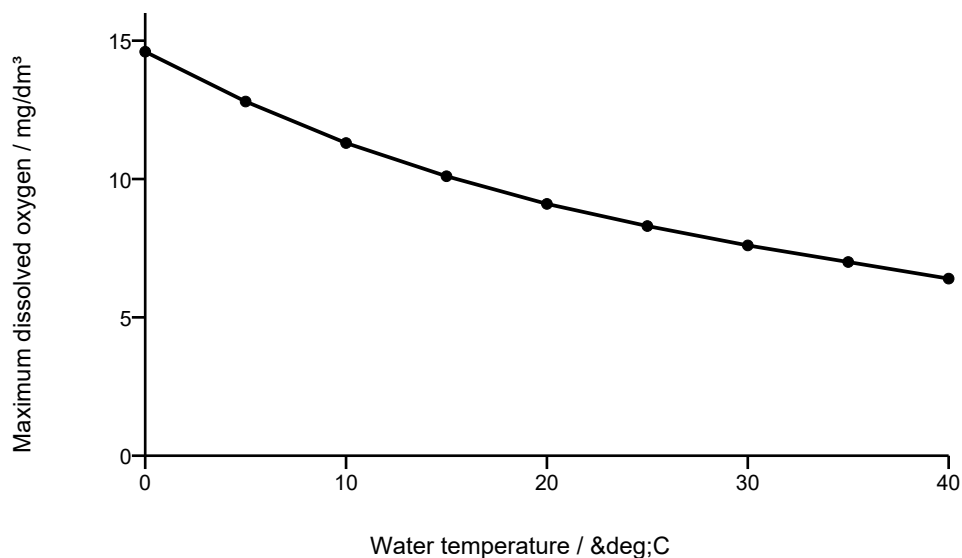
Questions by topic: Thermal & oil pollution - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1

Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg dm^{-3} .

Figure 1



0 1

. 1

Describe the relationship shown in Figure 1.

[1 mark]

0 1

. 2

Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

[1 mark]

Answer

°C

0 1

. 3

In summer the river upstream of the power station is at 20°C and the effluent raises the temperature by 7°C . Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

[2 marks]

Answer mg/dm³

0 1 . 4 Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1. **[2 marks]**

.....

.....

.....

.....

0 1 . 5 Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen. **[3 marks]**

1

.....

2

.....

3

.....

0 1 . 6 Describe how a cooling tower reduces the temperature of power station effluent water. **[1 mark]**

.....

.....

10

0 2 . 1

Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

1

.....

2

.....

3

.....

0 2 . 2

Explain how an oil spill at sea can kill seabirds.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 3

Explain **two** ways in which an oil slick can harm organisms that live in the water beneath it.

[2 marks]

.....

.....

.....

.....

0 2 . 4

Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

.....

.....

.....

.....

10

03 Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

03 . 1 Complete Table 1 by explaining how each feature reduces oil pollution. [5 marks]

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

03 . 2 Explain how each of the following reduces the risk of tanker accidents. [2 marks]

Offshore shipping routes

GPS / Automatic Identification System

0 3 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank. **[2 marks]**

0 3 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park. **[1 mark]**

10

0 4 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 4 . 1 Explain why inflatable booms were deployed immediately, and give **one** limitation of booms. **[3 marks]**

0 4 . 2 Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of. **[3 marks]**

0 4 . 3

Explain why dispersants were **not** used in the lagoon.

[2 marks]

0 4 . 4

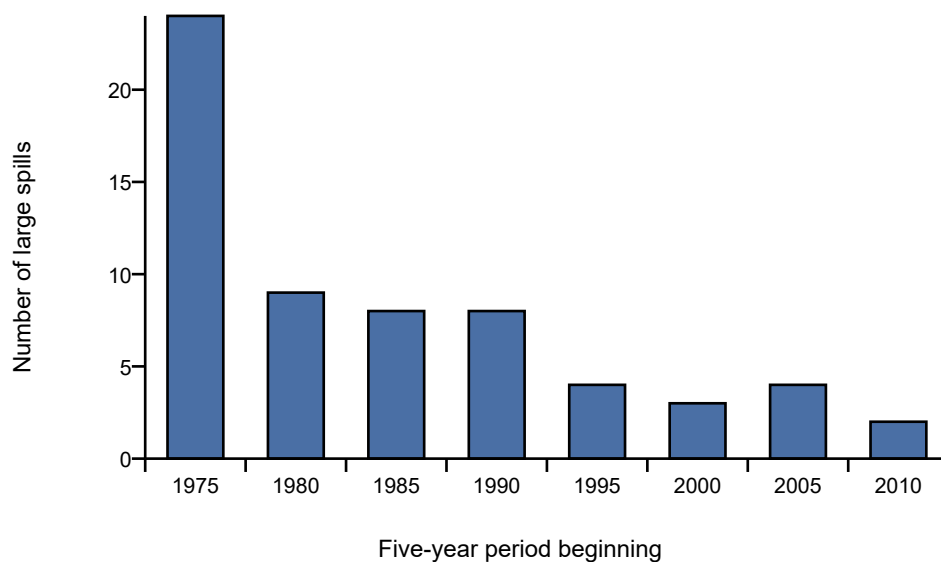
Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

10

0 5

Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1**Table 1**

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 5**1**

Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]**Answer**

%**0 5****2**

Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

.....

.....

.....

0 5 . 3

Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

1

2

3

0 5 . 4

Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

10

0 6 . 1

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops

☐

B Steam washing

☐

C Inflatable booms

☐

D Polymerising agents

☐

0 6 . 2

Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

.....

.....
.....
.....
06 . 3 Explain why deoxygenation is a more common problem in water than in air.
[2 marks]

.....
.....
.....
.....

5

END OF QUESTIONS